Preface

Blockchain technology has been the most promising alternative in recent years as far as distributed processing and storage of data are concerned. This volume presents the most recent theories and applications on the adjustment and execution of blockchain technology, progressively issue-based logical examination applications. The editors offer the fast headways in architecture, brilliant urban communities, digital physical frameworks, and web of things field. Blockchain-based information fortune in distributed computing and smart transportation frameworks are the main parts of this book. A few use cases with interruption identification frameworks, unmanned aeronautical vehicles, and online computer games are likewise fused. As blockchain for electronically casting a ballot framework and computerized right administration are moderate innovations and are utilized in numerous segments for dependable, savvy, and quick business exchanges, this book is an invited expansion on existing information. This book will address the engineering, plan objectives, difficulties, constraints, and potential answers for the blockchain-based notoriety frameworks. This book additionally joins in the examination of sanitation. It also focuses on the blockchain-based design perspectives of these intelligent architectures for evaluating and interpreting real-world trends. The chapters expand on different models that have shown considerable success in dealing with an extensive range of applications, despite their ability to extract complex hidden features and learn efficient representation in unsupervised environments for blockchain security pattern analysis.

The volume comprises 11 well-versed contributory chapters in different areas of blockchain technology.

Chapter 1 introduces the underlying technology, known as blockchain, of famous cryptocurrencies like Bitcoin. It starts by introducing the term blockchain, its analogies and characteristics. It then contains topics like history of blockchain and benefits of blockchain over traditional technologies. It emphasizes architecture of blockchain and various consensus mechanisms. The latter part of this chapter explores types of blockchain and the blockchain technology stack. It then lists various top blockchains used in contemporary days. It also explains various challenges of using blockchain such as energy consumption, scalability, public perception, etc. This chapter also explores the different types of attacks on blockchain.

The early days of cyberspace expansion saw many people become hesitant to use the internet facility due to two dominant reasons: obscurity and skepticism. The internet soon became a one-stop solution to a diverse range of sectors; a need to resolve issues such as data confidentiality and integrity gained prime importance. In 2008, an online currency called Bitcoin was launched by an anonymous developer or developer community called Satoshi Nakamoto, which was deemed to be a viable alternative to fiat or government-issued currency. What made Bitcoin famous and easily adaptable was the ease of accessibility in the form of a device application or simply "wallet" that included the nodes or "blocks" on which transaction data would be stored and processed instantaneously, and the intention behind it was to create a trustworthy cash system that would give individuals complete ownership of their transactions, not at risk of security breaches. With the advent and success of Bitcoin, new doors opened for other cryptocurrencies and digital assets like Ethereum and Litecoin, and it was observed that unlike fiduciary money, the sustainability of a thorough and transparent record must be maintained in connection with the exchange of digitalized money between any two parties whose anonymity would be preserved. Eventually, this became a turning point and marked the beginning of blockchain technology, which functioned as a distributed ledger system, ensuring secure transactions given its strong and complex cryptographic background strengthened with the help of hashing techniques and timestamps. Chapter 2 delves into this facet of blockchain technology in regard to cyber security.

A cyber-physical system (CPS) is a computer system that integrates real-world objects with embedded technologies to control and monitor physical processes. On the other hand, blockchain technology is a distributed, decentralized framework, which is the core concept behind Bitcoin. Blockchain technology can improve the reliability, security, and robustness of CPS-enabled critical infrastructure systems. Chapter 3 begins with a detailed overview of CPS and blockchain technology followed by a discussion on the various existing use cases of CPS applications.

Chapter 4 relates a new centralized ledger technology with a centralized validation process. It offers a single platform for all categories of real-time transactions and validations, unlike existing conventional blockchain technology. It offers three levels of hashing placed at the generator, server, and validator end for data security from data tampering and two levels of encryption for communication lines between generator-server and server-validator for packet security. This system ensures trustworthiness, authenticity, and CIA (confidentiality, integrity, and availability) to its end users while being real time in execution. The proposed system does not follow a chain-based file architecture. Due to this, no concept of chain break arises, and the problems that arise as a result of chain break in the blockchain are avoided.

Blockchain is a decentralized core architecture, which uses cryptographic validation, also known as hashing function, to link blocks with one another, and this linking plays a major role in ensuring the security, completeness, reliability, and accuracy of the data being stored and transparency in processing. Blockchain also provides a cryptographically safe and secure technique to obtain verified and immutable records in a chain that is chronologically ordered by discrete time stamps. These days, with almost everything becoming digital, the increasing rates of cyber threats, incidents, and policy violations have become a huge concern for organizations. It is only reasonable to use the vast variety of features that are offered by blockchain to assist intrusion detection systems (IDSs). IDSs are mainly used to monitor networks for malicious activities and report such activities. In Chapter 5, the authors discuss the different IDSs, the current situation of IDSs, and how to enhance the IDSs using blockchain.

The internet of things (IoT) has expanded the reachability of pervasive applications in different domains. Smart healthcare is one such area where IoT-based approaches have been successfully demonstrated in the recent past. However, issues related to the scaling of large ad-hoc elements is still a major challenge. Medical delivery drones have come up as a new variant of unmanned aerial vehicle (UAV) to pave holistic orientation into the smart health service paradigm. Lack of augmentation aware notions has resisted the realization of an actual intervention of IoT-based medical delivery via UAVs. Proper architectural frameworks and models are needed to efficiently integrate IoT with the medical delivery drones. However, major security drawbacks should be investigated a priori for health delivery services in coming days. One can use the blockchain-centric approaches to fill this gap in the IoT-based medical delivery drone ecosystem. In Chapter 6, the authors firstly present the details about the drone suppliers and perform a comparative analysis of such drones for medical delivery. Secondly, a state of the art on blockchain-assisted IoT-based drone-centric medical delivery is also presented. Finally, some of the key challenges associated with existing drone delivery mechanisms are highlighted.

A huge majority of digital media in today's age is consumed over the internet. Any digital content uploaded online is extremely vulnerable to leakage through piracy, illegal spread, and sharing of digital copies of files to users who have not paid for the content's consumption. Many times, authentication of content to trace back its source of spread fails due to the anonymity of the said source. The current system of centralized content distribution networks working in conjunction with digital rights management systems to provide and authenticate users consuming digital content is faulty and can be replaced with a far more robust peer-to-peer (P2P) system that operates on blockchain technology. Such a system is illustrated in Chapter 7, which would benefit consumers since content will no longer be held by a centralized server but rather be decentralized geographically, ensuring availability through physical hindrances such as natural disasters or digital blockages such as distributed denial of service attacks, etc. The system will also help digital content creators of various fields such as photography, music, etc., by providing the ability to detect exact points of content breaches as well as maintaining a chain of transactions in case needed in a future investigation.

The past decade has envisaged a paradigmatic shift in the digitalization of executing and recording financial transactions by developing intelligent algorithms secured through cryptography. The architecture of blockchain technology holds tremendous potential for metamorphosing various industrial sectors like healthcare, life sciences, and clinical and medical data management. In recent years, there has been a continuous surge for adoption in the biomanufacturing sector by enhancing supply chain management and customer relationship through blockchain technology. It confers substantial impact on bioprocess design, operational efficiency, reduced maintenance downtime, management of spare part inventory, and continuous online monitoring that has paved the way for inevitable transformation in the biomanufacturing sector. In Chapter 8, the authors highlight the role of blockchain technology in the biomanufacturing sector by highlighting its implementation on various unit operations of the biomanufacturing sector. The technology is still in its nascent stage, but it can be wooed at a global scale for wider applicability to make the biomanufacturing sector more sustainable.

The concept of voting has been in use for many purposes in the past decades. The conventional ballot-based voting (paper and ballot box) approach has many inherent use cases that could lead to severe malpractices and manipulations during and after the voting process. In addition, it would also lead to many debatable situations where the physical voting-related practices could be questionable. To overcome the issues raised in the conventional voting system, an electronic voting (e-voting) system came into being. Although the e-voting approaches are efficient and address the drawbacks of conventional e-voting systems, they raise new challenges to researchers, such as voter identity verification, voter privacy protection, vote verifiability and integrity, and so on. Since blockchain guarantees data security, it can protect the system from coercion attacks so can provide better solutions for transparent and trusted e-voting systems. In Chapter 9, the authors explore various e-voting protocols and present a comparative study of performance, security features, and limitations. Further, they focus on the blockchain-based e-voting protocols to discuss the major achievements and challenges in the blockchain-based e-voting systems.

Blockchain technology is a decentralized distributed ledger that maintains a list of transactions in a P2P network in which the security is provided using cryptographic techniques. The history is recorded in every block, and these are cryptographically connected and secured over time. Moreover, this technology has been used for maintaining cryptocurrencies, digital contracts, land ownership, public records, etc., so far. Future applications are likely to include education, intellectual property, supply chain management, medicine, and science. Also, capital expenditures in blockchain technology are expected to hit USD 13.96 billion by 2022. A blockchain-based approach is presented in Chapter 10 to serve as a perfect solution in which the data security and privacy protection are the major concerns in developing a secure drug supply chain. The performance measures such as central dependency, security, smart contract management, encryption, error rate, user privacy, and traceability are considered for the comparison of the proposed technique with the existing approaches.

Artificial intelligence (AI) and blockchain, these two technologies have recently been the trendiest and most revolutionary progressive technologies. Blockchain technology can automate payments in cryptocurrency and provide admittance in a decentralized, safe, and trustworthy way to shared records, transactions, and log ledgers. Also, with smart contracts, blockchain can handle interactions between participants without an interceder. Furthermore, AI provides intelligence and smart decision-making for human-like machines. Chapter 11 presents a thorough analysis of AI and blockchain integration ability, possibilities, and applications. The incorporation of AI was evaluated, and blockchain will influence industry 4.0. We also recognize and address the open issues of AI and blockchain fusion.

This volume will attract researchers, practitioners, academics, and industry professionals related to design, impact, challenges, solutions, and emerging applications of the blockchain in various domains, thereby serving as an excellent resource for the following areas: smart cities, CPSs, cloud computing, intelligent transportation systems, network security, video gaming, IoT, supply chain, reputation, and financial technology.

September 2021

SK Hafizul Islam
Kalyani, India

Arup Kumar Pal
Dhanbad, India

Debabrata Samanta
Bangalore, India

Siddhartha Bhattacharyya
Birbhum, India